SIX-TERM EXACT SEQUENCE FOR COMPLETED NAIVE COTANGENT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 742-777, following the statement at 724-740.

Editorial note. No omissions flagged; the 'reader verifies' phrase covers the assembly of the displayed diagram whose rows are exact by inspection.

Editorial note. The source typo 'Combinging' (line 754) is preserved verbatim.

Allowed standard prerequisites (editorial).

- Well-definedness of the completed NL in K(B). lemma-NL-up-to-homotopy (carded L26281, cited at ~743-745). Makes the tensor product NL^wedge_{B/A} tensor_B C meaningful.
- Noetherianness of topologically finite type rings. lemma-topologically-finite-type-Noetherian (same chapter, cited at ~746-747). (B, IB) is a Noetherian pair, so C sits in the corresponding category.
- Snake lemma. Algebra, Lemma algebra-lemma-snake (cross-chapter, cited at ~775-776). Extracts the six-term exact sequence from the displayed diagram.

Source reference. Original reference 07JW: focused-seeds/geometry/S9-restricted-exact-sequence-NL/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GAE: sources/stacks-restricted/pinned-original-source.tex, source lines 506–506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AJQ: sources/stacks-restricted/pinned-original-source.tex, source lines 395–395. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let A be a Noetherian ring and let $I \subset A$ be an ideal. Let $B \to C$ be morphism of (0AL4). Then there is an exact sequence*

$$\begin{array}{l} C \otimes_B H^0(NL^\wedge_{B/A}) \longrightarrow H^0(NL^\wedge_{C/A}) \longrightarrow H^0(NL^\wedge_{C/B}) \longrightarrow 0 \\ \\ H^{-1}(NL^\wedge_{B/A} \otimes_B C) \longrightarrow H^{-1}(NL^\wedge_{C/A}) \longrightarrow H^{-1}(NL^\wedge_{C/B}) \end{array}$$

See proof for elucidation.

Proof. Observe that taking the tensor product $NL^\wedge_{B/A} \otimes_B C$ makes sense as $NL^\wedge_{B/A}$ is well defined up to homotopy by Lemma 0GAE. Also, (B, IB) is pair where B is a Noetherian ring (Lemma 0AJQ) and C is in the corresponding category (0AL4). Thus all the terms in the 6-term sequence are (well) defined.

Choose a presentation $B = A[x_1, \ldots, x_r]^\wedge/J$. Choose a presentation $C = B[y_1, \ldots, y_s]^\wedge/J'$. Combinging these presentations gives a presentation

$$C = A[x_1, \ldots, x_r, y_1, \ldots, y_s]^\wedge/K$$

Then the reader verifies that we obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \bigoplus C\mathrm{d}x_i & \longrightarrow & \bigoplus C\mathrm{d}x_i \oplus \bigoplus C\mathrm{d}y_j & \longrightarrow & \bigoplus C\mathrm{d}y_j & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ & & J/J^2 \otimes_B C & \longrightarrow & K/K^2 & \longrightarrow & J'/(J')^2 & \longrightarrow & 0 \end{array}$$

with exact rows. Note that the vertical arrow on the left hand side is the tensor product of the arrow defining $NL^\wedge_{B/A}$ with id_C. The lemma follows by applying the snake lemma (Algebra, Lemma 07JW). □